Lumentra CRM User Guide

Lumentra AI LLC

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1. CRM Overview

What Is the Lumentra CRM?

The Lumentra CRM is a built-in customer relationship management system designed specifically for businesses that use Lumentra's voice AI platform. It provides a unified view of every customer interaction – from the first phone call through booking, follow-up, and long-term engagement – without requiring a separate CRM product.

Unlike standalone CRM systems that require manual data entry and integration work, the Lumentra CRM is deeply woven into the voice agent infrastructure. Every call, every booking, and every customer interaction automatically flows into the CRM, creating a real-time picture of customer engagement without any manual effort from business owners or staff.

The CRM supports multi-tenant isolation, meaning each business using the Lumentra platform has its own private database partition with no data leakage between tenants. All queries, filters, and automation rules operate within the boundaries of a single tenant.

How the CRM Integrates with Voice Calls

Every inbound phone call handled by the Lumentra voice agent feeds directly into the CRM:

1. **Automatic Contact Creation.** When a caller's phone number is not recognized, the system creates a new contact record automatically. If the caller is already in the database, their existing record is updated with the new interaction. The lookup uses a high-performance cache with a target response time under 50 milliseconds.
2. **Post-Call Automation.** After every call is logged, the system runs a set of automation rules that can create deals, generate follow-up tasks, update lead statuses, and even upgrade contacts to VIP status – all without manual intervention. These rules execute asynchronously so they never block the call logging response.
3. **Real-Time Engagement Scoring.** Each call, booking, and interaction contributes to a contact's engagement score. This score is recalculated based on recency, frequency, conversion rate, loyalty, and lifetime value. The scoring algorithm uses six weighted factors that sum to a maximum of 100 points.
4. **Linked Records.** Calls are linked to contacts, bookings, deals, and tasks. Clicking into any call shows its full context: who called, what they needed, whether a booking was made, and what follow-up is required.
5. **Sentiment Analysis.** Every call receives a sentiment score between -1.0 (very negative) and +1.0 (very positive). This feeds into the contact's sentiment average, helping identify customers who may be at risk of churning.
6. **Intent Detection.** The voice agent detects and records caller intents (such as "booking", "hours_inquiry", "pricing", "cancellation") which are stored on the call record and available for reporting and segmentation.

Core CRM Modules

Module	Purpose
Contacts	Customer database with metrics
Deals	Sales pipeline with Kanban board
Tasks	Follow-ups and action items
Call History	Call log with transcripts
Analytics	Charts and conversion metrics
Notifications	Booking confirmations and reminders

Who Should Use This Guide

This guide is intended for:

- **Business Owners** who want to understand how their customer data is organized and how to leverage the CRM for growth.
- **Office Managers and Receptionists** who handle day-to-day contact management, deal tracking, and task completion.
- **Administrators** who configure the CRM settings, notification templates, and pipeline stages for their business.
- **Developers** who integrate with the CRM via the REST API or build custom workflows on top of the platform.

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2. CRM Architecture and Data Model

Entity Relationship Overview

The Lumentra CRM is built around five core entities that are interconnected through foreign key relationships. Understanding how these entities relate to each other is essential for effective use of the system.

Core Entities:

- **Contacts** – The central entity. Every person who calls or interacts with your business has a contact record.
- **Calls** – Every phone call handled by the voice agent is stored as a call record, linked to a contact.
- **Bookings** – Appointments, reservations, or service orders created during calls, linked to both a contact and a call.
- **Deals** – Business opportunities that move through your sales pipeline, linked to a contact and optionally a call.
- **Tasks** – Follow-up items and action items, linked to a contact, a deal, and/or a call.

Supporting Entities:

- **Contact Notes** – Free-form text notes attached to contacts, with type classification and pinning.
- **Contact Activity** – A chronological audit log of every action taken on a contact.
- **Notifications** – Messages sent to contacts via email, SMS, or push.
- **Notification Templates** – Configurable message templates with variable substitution.
- **Resources** – Staff, rooms, equipment, or services used in scheduling.
- **Availability Slots** – Time slots available for booking, linked to resources.

Data Flow: From Call to CRM

The following sequence describes how data flows through the system when a customer calls:

1. **Inbound call arrives** via SignalWire SIP trunk, routed through LiveKit SIP to the voice agent.
2. **Phone lookup** – The agent calls the internal API endpoint `POST /api/contacts/find-or-create` with the caller's phone number. The system checks the in-memory cache first (sub-50ms), then falls back to a database query.
3. **Contact record resolved** – Either an existing contact is found and returned, or a new contact is created with source set to "call", status "active", and lead_status "new".
4. **Call proceeds** – The voice agent conducts the conversation, detecting intents and recording the transcript.
5. **Call ends** – The agent logs the call via the internal API with full metadata: duration, outcome, transcript, summary, sentiment, and intents.
6. **Post-call automation fires** – Five automation rules are evaluated sequentially (see Section 14). These can create deals, tasks, and update contact status.
7. **Engagement score recalculated** – The contact's engagement score is updated based on the new call data.
8. **Notifications triggered** – If a booking was made, confirmation notifications are queued for delivery.

Multi-Tenant Isolation

Every entity in the CRM includes a `tenant_id` column that acts as a partition key. The system enforces tenant isolation at multiple levels:

- **Database Level:** PostgreSQL Row-Level Security (RLS) policies ensure that queries can only return data belonging to the authenticated tenant.
- **Application Level:** Every API route extracts the tenant ID from the authenticated session and includes it as a WHERE clause condition. This provides defense-in-depth even if RLS is bypassed.
- **Cache Level:** The contact lookup cache is keyed by both tenant ID and phone number, preventing cross-tenant cache leakage.

Database Tables Summary

Table	Row Count Typical	Primary Key
contacts	100 - 50,000 per tenant	UUID
calls	500 - 200,000 per tenant	UUID
bookings	100 - 100,000 per tenant	UUID
deals	50 - 10,000 per tenant	UUID
tasks	50 - 5,000 per tenant	UUID
contact_notes	Varies	UUID
contact_activity	Varies	UUID
notifications	Varies	UUID
notification_templates	6-20 per tenant	UUID

Foreign Key Relationships

- `calls.contact_id` references `contacts.id`
- `calls.booking_id` references `bookings.id`
- `bookings.contact_id` references `contacts.id`
- `bookings.call_id` references `calls.id`
- `deals.contact_id` references `contacts.id`
- `deals.call_id` references `calls.id`
- `tasks.contact_id` references `contacts.id`
- `tasks.deal_id` references `deals.id`
- `tasks.call_id` references `calls.id`
- `contact_notes.contact_id` references `contacts.id`
- `contact_notes.call_id` references `calls.id`
- `contact_notes.booking_id` references `bookings.id`
- `contact_activity.contact_id` references `contacts.id`
- `notifications.contact_id` references `contacts.id`
- `notifications.booking_id` references `bookings.id`
- `notifications.call_id` references `calls.id`

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3. Contacts Management

Navigating to Contacts

Open the dashboard and select **Contacts** from the sidebar navigation. The contacts list displays all customers in your database with summary information in a sortable, searchable table.

Contacts List View

The contacts table shows the following columns:

Column	Description
Name	Contact name and phone number
Email	Email address or a dash
Status	Badge: Active, Inactive, etc.
Bookings	Total booking count
Calls	Total call count
Score	Engagement score (0-100)
Last Contact	Date of most recent interaction

The “Bookings” column header adapts to your industry terminology. For restaurants it shows “Reservations”, for salons it shows “Appointments”, and so on.

The engagement score column is color-coded:

- **Green (80+):** Highly engaged, VIP-level customers
- **Amber (50-79):** Moderately engaged customers
- **Gray (below 50):** Low engagement, may need re-engagement

Searching Contacts

Use the search bar at the top of the page to find contacts by name, email, or phone number. The search is case-insensitive and applies a 300ms debounce to avoid excessive queries while you type.

The search operates as an ILIKE pattern match, meaning partial matches work. For example, searching for “john” will match “John Smith”, “Johnson”, and “john@example.com”.

Filtering Contacts

Click the **Filters** button to narrow the contacts list. Available filters include:

- **Status:** active, inactive, blocked, vip, churned (multi-select)
- **Lead Status:** new, contacted, qualified, converted, lost (multi-select)
- **Source:** call, booking, import, manual, sms, web (multi-select)
- **Tags:** filter by one or more tags
- **Has Bookings:** yes/no
- **Has Calls:** yes/no
- **Date Ranges:** created_after, created_before, last_contact_after, last_contact_before

Filters can be combined. For example, you can filter for VIP contacts who were created after January 2026 and have made at least one booking.

Sorting

Click any sortable column header (Name, Bookings, Calls, Score, Last Contact) to sort ascending. Click again to reverse the sort order. The default sort is by `last_contact_at` in descending order, showing the most recently active contacts first.

Pagination

The contacts list loads 20 records per page by default. Use the pagination controls at the bottom of the table to navigate between pages or change the number of records per page. The API supports a maximum of 100 records per page.

Selecting Multiple Contacts

Check the checkbox next to individual contacts, or use the header checkbox to select all visible contacts. When contacts are selected, a selection indicator appears showing the count and a **Clear** button.

Bulk operations available for selected contacts:

- **Bulk Tag:** Add a tag to all selected contacts at once.

Creating a Contact

1. Click the **Add Contact** button in the top-right corner.
2. Fill in the contact form:
    - **Phone Number** (required): The contact’s phone number. This field cannot be changed after creation. The system normalizes the phone number to E.164 format automatically.

    - **First Name** (optional)
    - **Last Name** (optional)
    - **Email** (optional): Must be a valid email format. Stored in lowercase.
    - **Company** (optional)
    - **Notes** (optional): Free-text notes about this contact.
3. Click **Add Contact** to save.

New contacts are created with:

- Status: `active`
- Lead Status: `new`
- Source: `manual`
- Engagement Score: `0`
- All metric counters: `0`
- Preferred Language: `en`

If a contact with the same phone number already exists for your tenant, the system returns a 409 Conflict error.

Viewing Contact Details

Click any row in the contacts table to open the contact detail drawer on the right side of the screen. The detail view includes:

Profile Section:

- Avatar initial, full name, and company
- Tags displayed as badges
- Phone number, email address, and company
- Three stat cards: Bookings count, Calls count, and Engagement Score
- Lifetime value (if greater than $0), displayed in a highlighted card

Tabs:

- **Activity** – Chronological timeline of all interactions (calls, bookings, SMS, notes, status changes)
- **Notes** – All notes attached to the contact, with pinned notes shown first
- **Details** – Full contact information including status, lead status, source, first/last contact dates, preferred language, preferred contact method, and communication preferences

Editing a Contact

1. Open the contact detail drawer.
2. Click the **Edit** (pencil) icon in the drawer header.
3. Modify the fields you want to update:
    - Email, first name, last name, company, notes
    - Status: active, inactive, blocked, vip, churned
    - Lead Status: new, contacted, qualified, converted, lost
    - Preferred Contact Method: phone, sms, email
    - Preferred Contact Time (free text, e.g., “mornings”)
    - Preferred Language (ISO 639-1 code, e.g., “en”, “es”)
    - Timezone (IANA format, e.g., “America/New_York”)
    - Communication Preferences: do_not_call, do_not_sms, do_not_email, marketing_opt_in
    - Tags (array of strings)
    - Custom Fields (JSON object for business-specific data)
4. Click **Save Changes**.

Note: The phone number cannot be changed after the contact is created. If a contact’s phone number needs to change, create a new contact and use the merge feature.

Deleting a Contact

1. Open the contact detail drawer.
2. Click the **Delete** (trash) icon in the drawer header.
3. Confirm the deletion when prompted.

Deletion is a soft delete – the contact's status is set to `inactive` rather than being permanently removed. All linked calls, bookings, and notes are preserved. The soft delete also logs a `status_changed` activity entry for audit purposes.

Communication Preferences

Each contact has four compliance-related flags:

Flag	Default	Meaning
`do_not_call`	false	Opted out of phone calls
`do_not_sms`	false	Opted out of SMS messages
`do_not_email`	false	Opted out of email
`marketing_opt_in`	false	Consented to marketing

These flags are shown in the Details tab with green ("OK") or red ("No") indicators. When `marketing_opt_in` is set to true, the system records the timestamp in `marketing_opt_in_at`.

The notification system respects these flags. If `do_not_sms` is true, SMS notifications will be skipped for that contact. If `do_not_email` is true, email notifications will be skipped. Marketing-type notifications require `marketing_opt_in` to be true.

Tags

Tags are free-form text labels attached to contacts. They appear as badges in the contact detail view. Tags are useful for segmentation, filtering, and organizing contacts into groups.

Common Tag Examples:

- "vip-guest", "repeat-customer", "event-booking"
- "insurance-verified", "new-patient", "referral"
- "large-party", "dietary-restrictions", "special-occasion"
- "fleet-account", "warranty-active", "commercial"

Adding Tags:

- Edit the contact and update the tags array, or
- Use the API endpoint `PATCH /api/contacts/:id/tags` with `{ "add": ["tag1", "tag2"] }`

Removing Tags:

- Use `PATCH /api/contacts/:id/tags` with `{ "remove": ["tag1"] }`

Bulk Tagging:

- Select multiple contacts in the table, then use the bulk tag operation to add a tag to all selected contacts. This calls `POST /api/contacts/bulk/tags` with the list of contact IDs and the tag to add.

Contact Notes

Notes provide a way to record observations, preferences, and follow-up items against a contact.

Note Types:

Type	Use Case
`general`	General-purpose notes
`call_summary`	Auto-generated call summary
`booking_note`	Notes about a booking
`preference`	Customer preferences
`complaint`	Customer complaints
`compliment`	Positive feedback
`follow_up`	Items needing follow-up
`internal`	Internal-only notes
`system`	System-generated notes

Notes can be **pinned** to appear at the top of the list, and marked as **private** to restrict visibility. Each note records the author's name and creation timestamp.

Notes are ordered with pinned notes first, then by creation date descending.

Merging Duplicate Contacts

When the same customer has multiple contact records (for example, they called from two different phone numbers), you can merge them:

1. Identify the primary contact (the record you want to keep).
2. Identify the secondary contacts (the duplicates to merge into the primary).
3. Call `POST /api/contacts/merge` with `primary_id` and `secondary_ids`.

The merge operation performs the following steps:

- Moves all calls from secondary contacts to the primary contact
- Moves all bookings from secondary contacts to the primary contact
- Moves all notes from secondary contacts to the primary contact
- Moves all activity history from secondary contacts to the primary contact
- Combines metric counters (total_calls, total_bookings, lifetime_value_cents, etc.) by summing them
- Merges tags (union of all tags, no duplicates)
- Uses the earliest `first_contact_at` and the latest `last_contact_at`
- Soft-deletes secondary contacts (sets status to `inactive`)
- Logs a `merged` activity entry on the primary contact
- Invalidates the cache for all affected contacts

---

4. Contact Lifecycle Management

From Anonymous Caller to VIP

Every contact in the Lumentra CRM follows a natural lifecycle that progresses from first interaction to long-term customer relationship. Understanding this lifecycle helps businesses identify opportunities for engagement at each stage.

Stage 1: Anonymous Caller

When someone calls your business for the first time, the voice agent performs a phone lookup. If the number is not found, a new contact is created with minimal information:

- Phone number (the only guaranteed field)
- Source: "call"
- Status: "active"
- Lead Status: "new"
- Engagement Score: 0

At this point, the contact is essentially anonymous – just a phone number.

Stage 2: Identified Contact

During the call, the voice agent may collect the caller's name. If the caller provides their name, it is stored on the contact record. After the call, the post-call automation may also update the contact based on the call outcome. The contact now has:

- Name (first and/or last)
- At least one call record with transcript and summary
- A detected intent (inquiry, booking, support, etc.)
- An initial engagement score (typically 15-25 for a single call)

Stage 3: Qualified Lead

After multiple interactions or a booking, the contact becomes a qualified lead. Indicators include:

- Lead status progressed to "contacted" or "qualified"
- Multiple calls in the activity history
- Possibly a deal created in the pipeline
- Engagement score in the 30-59 range (warm)
- Tags added based on interests or needs

Stage 4: Converted Customer

When the contact completes a booking or purchase, the system automatically:

- Updates lead status to "converted"
- Creates a deal at the completed/won stage
- Increments the lifetime value counter
- Boosts the engagement score via conversion rate and booking frequency factors

Stage 5: Repeat Customer

Repeat customers show up through increasing metric counters:

- Multiple completed bookings
- Rising engagement score (60-79 range, "hot" level)
- Consistent call frequency
- Growing lifetime value

Stage 6: VIP Status

The system automatically upgrades contacts to VIP status when they meet all three criteria:

- 3 or more total calls
- Engagement score of 80 or higher
- Current status is not already VIP

VIP contacts receive:

- A distinct badge in the contacts list
- Priority consideration in engagement-level queries
- VIP-level engagement classification regardless of score drops (once the status is set manually, it persists unless explicitly changed)

Stage 7: Churned Contact

Contacts who stop engaging eventually see their engagement score decay through the recency factor. If a contact has not interacted in over 90 days, their recency score drops to 0, significantly reducing their overall engagement score. Business owners can manually set the status to "churned" to flag these contacts for re-engagement campaigns.

Status State Machine

Contact status transitions follow these rules:

- **active** – The default state for all new contacts. Can transition to any other status.
- **inactive** – Set when a contact is soft-deleted or merged as a secondary. Can be reactivated to "active".
- **blocked** – Set manually when a contact should be excluded from all communications. Can be unblocked to "active".
- **vip** – Set automatically by post-call automation or manually by staff. Can transition to any other status.
- **churned** – Set manually to flag disengaged contacts. Can be reactivated to "active" if the contact re-engages.

Lead Status Funnel

The lead status is separate from the contact status and tracks the sales/conversion journey:

```
New --> Contacted --> Qualified --> Converted
                                       |
                                     Lost
```

Lead status transitions are mostly manual, with two automatic transitions:

- **new** is set automatically when a first-time caller makes an inquiry
- **converted** is set automatically when a call results in a booking

---

5. Lead Tracking

Lead Status Flow

Every contact has an optional `lead_status` field that tracks where they are in the customer journey. The five lead statuses form a funnel:

```
New --> Contacted --> Qualified --> Converted
                                       |
                                     Lost
```

Status	Description
new	First-time caller or newly imported
contacted	Business has reached out
qualified	Evaluated as a genuine prospect
converted	Completed a booking or purchase
lost	Did not convert, no longer pursued

Automatic Lead Status Updates

The post-call automation system updates lead status in two scenarios:

1. **Booking outcome:** When a call results in a booking, the contact's lead_status is automatically set to `converted`.
2. **First-time inquiry:** When a first-time caller's call is classified as an `inquiry`, the lead_status is confirmed as `new` (the default).

All other transitions (contacted, qualified, lost) are manual – update them through the contact edit form or the API.

Best Practices for Lead Management

Daily Review Process:

1. Filter contacts by lead_status = "new" to see fresh leads

2. Review the call summary and intents for each new lead
3. Move promising leads to “contacted” after reaching out
4. Mark unresponsive or unqualified leads as “lost”

Weekly Pipeline Review:

1. Filter by lead_status = “qualified” to review the sales pipeline
2. Check the engagement score trend – is it rising or falling?
3. Create follow-up tasks for stalled leads
4. Review the conversion rate metric on the analytics page

Re-Engagement Strategy:

1. Filter by lead_status = “lost” and last_contact_after a recent date
2. Review the call transcripts to understand why they were lost
3. Consider reaching out with a special offer or updated information
4. If re-engaged, move back to “contacted”

---

6. Engagement Scoring Deep Dive

How the Engagement Score Works

Each contact has an `engagement_score` from 0 to 100 that quantifies how engaged they are with your business. The score is calculated from six weighted factors that sum to exactly 100 maximum points.

The score is recalculated:

- After every call is logged (via post-call automation)
- When a booking status changes
- When explicitly triggered via the API (`POST /api/contacts/:id/recalculate-score`)
- During batch recalculation operations

Scoring Factors and Weights

Factor	Weight	Max Contribution
Recency	25	25 points
Booking Frequency	20	20 points
Call Frequency	15	15 points
Conversion Rate	15	15 points
Loyalty	15	15 points
Lifetime Value	10	10 points

Factor 1: Recency (25 points)

The recency factor measures how recently the contact last interacted with your business. It uses the `last_contact_at` timestamp, which is updated whenever a call, booking, or other interaction occurs.

Recency Score Thresholds:

Days Since Last Contact	Raw Score
0-7 days	100
8-14 days	80
15-30 days	60
31-60 days	40
61-90 days	20
91+ days	0

Contribution Calculation:

Contribution = (Raw Score * Weight) / 100

Example: Contact last seen 10 days ago -> Raw Score 80 -> Contribution = (80 * 25) / 100 = 20 points

Factor 2: Booking Frequency (20 points)

Measures the number of bookings in the last 90 days. Each booking adds 33 points to the raw score, capped at 100.

Bookings (90 days)	Raw Score
0	0
1	33
2	66
3+	100

Contribution Calculation:

Example: 2 bookings in 90 days -> Raw Score 66 -> Contribution = (66 * 20) / 100 = 13.2 points

Factor 3: Call Frequency (15 points)

Measures the number of calls in the last 30 days. Each call adds 25 points to the raw score, capped at 100.

Calls (30 days)	Raw Score
0	0
1	25
2	50
3	75
4+	100

Contribution Calculation:

Example: 3 calls in 30 days -> Raw Score 75 -> Contribution = (75 * 15) / 100 = 11.25 points

Factor 4: Conversion Rate (15 points)

Measures the percentage of recent calls that resulted in a booking outcome. The raw score is double the conversion percentage, capped at 100. This means a 50% or higher conversion rate achieves the maximum raw score.

Conversion Rate	Raw Score
0%	0
25%	50
50%+	100

Contribution Calculation:

Example: 2 out of 4 calls resulted in bookings (50%) -> Raw Score 100 -> Contribution = (100 * 15) / 100 = 15 points

Factor 5: Loyalty (15 points)

Measures the "show rate" – the percentage of total bookings that were completed (as opposed to cancelled or no-showed). A 100% show rate yields the maximum raw score.

Show Rate	Raw Score
0%	0
50%	50
100%	100

Contribution Calculation:

Example: 8 completed out of 10 total bookings (80%) -> Raw Score 80 -> Contribution = (80 * 15) / 100 = 12 points

Factor 6: Lifetime Value (10 points)

Measures the total revenue generated by this contact, stored as `lifetime_value_cents`. Uses tiered thresholds:

Lifetime Value	Raw Score
$0-$9.99	0
$10-$49.99	20
$50-$199.99	40
$200-$499.99	60
$500-$999.99	80
$1,000+	100

Contribution Calculation:

Example: $350 lifetime value -> Raw Score 60 -> Contribution = (60 * 10) / 100 = 6 points

Engagement Levels

The total score maps to four engagement levels:

Level	Score Range	Meaning
VIP	80-100	Highest-value customers
Hot	60-79	Highly engaged
Warm	30-59	Moderate engagement
Cold	0-29	Low or no engagement

A contact's engagement level is stored on the contact record as `engagement_level` and is updated every time the score is recalculated.

Special case: If a contact's status is manually set to "vip", their engagement level is always classified as "vip" regardless of score.

Score Decay Over Time

The engagement score naturally decays over time through the recency factor. If a contact stops interacting:

- After 7 days: Recency drops from 100 to 80 (-5 points contribution)

- After 14 days: Recency drops to 60 (-10 points total)
- After 30 days: Recency drops to 40 (-15 points total)
- After 60 days: Recency drops to 20 (-20 points total)
- After 90 days: Recency drops to 0 (-25 points total)

This means that even a previously highly-engaged customer will see their score drop by up to 25 points if they go silent for 3 months.

Viewing the Engagement Breakdown

To view the detailed factor breakdown for any contact, call:

`GET /api/contacts/:id/engagement`

The response includes:

- `score`: The total engagement score (0-100)
- `level`: The engagement level (cold, warm, hot, vip)
- `factors`: An array of six factor objects, each with:
  - `name`: Factor name (e.g., "Recency")
  - `value`: The raw metric value (e.g., 5 for days since last contact)
  - `weight`: The factor weight (e.g., 25)
  - `contribution`: The actual points contributed (e.g., 25.0)

Batch Recalculation

For administrative purposes, you can trigger a batch recalculation of all engagement scores for a tenant. This is useful after a data import or migration. The batch operation processes contacts sequentially and returns a summary of updated and failed records.

---

7. Deal Management

What Is a Deal?

A deal represents a business opportunity moving through your sales pipeline. Depending on your industry, deals may be called by different names:

Industry	Deal Label	Example Name
Default / Generic	Deal	"Website Redesign Project"
Medical / Dental	Case	"John Smith - Case"
Restaurant	Reservation	"Jane Doe - Reservation"
Hotel / Motel	Booking	"Guest Booking"
Salon	Appointment	"Haircut Appointment"
Auto Service	Service Order	"Oil Change Service Order"

Navigating to Deals

Select **Pipeline** from the sidebar navigation. The deals page displays a Kanban board showing all active deals organized by stage.

Pipeline Kanban Board

The pipeline is displayed as a horizontal Kanban board. Each column represents a stage, showing:

- Stage name and color indicator
- Count of deals in that stage

- Total monetary value of deals in that stage
- Individual deal cards

The header shows the total number of deals and the total pipeline value (excluding terminal stages like Won/Lost/Completed).

Deal Cards

Each deal appears as a card in its stage column, displaying:

- **Deal name** (truncated if too long)
- **Contact name** (if linked to a contact)
- **Company** (if specified, with a building icon)
- **Amount** (formatted as currency, e.g., "$5,000")
- **Expected close date** (relative: "Today", "Tomorrow", "3d", or overdue with red text)

A drag handle appears on hover for drag-and-drop functionality.

Moving Deals Between Stages

To move a deal to a different stage, drag the deal card from one column and drop it into another. The update happens optimistically (the UI updates immediately) and syncs to the server in the background. If the server update fails, the card reverts to its original position.

The system validates that the target stage is valid for your industry pipeline. You cannot drag a deal to a stage that does not exist in your industry configuration. The validation is performed server-side via the `PATCH /api/deals/:id/stage` endpoint.

Creating a Deal

1. Click the **Add [Deal Label]** button in the top-right corner.
2. Fill in the deal form:
    - **Name** (required): A descriptive name for the deal.
    - **Company** (optional): The company associated with this deal.
    - **Amount ($)** (optional): The monetary value. Enter as dollars; stored internally as cents.
    - **Expected Close** (optional): The expected close date.
    - **Stage** (optional): The initial stage. Defaults to the first stage in your industry pipeline.
    - **Contact** (optional): Link to an existing contact.
    - **Description** (optional): Additional details.
3. Click **Add [Deal Label]** to save.

Editing a Deal

Open the deal form by clicking a deal card. Update the fields and click **Save Changes**. You can change the name, company, amount, expected close date, stage, contact link, and description.

Archiving a Deal

Deals are soft-deleted (archived) rather than permanently removed. An archived deal sets the `archived_at` timestamp and is excluded from pipeline views and searches. To archive a deal, use `DELETE /api/deals/:id`.

Automatic Deal Creation

The post-call automation creates deals automatically in two scenarios:

1. **Booking outcome:** When a call results in a booking, a deal is created at the completed/won stage with source set to `call`. Example: "John Smith - Case" (for medical) or "Jane Doe - Reservation" (for restaurant).

2. **First-time inquiry:** When a first-time caller's call is classified as an `inquiry`, a deal is created at the default/initial stage with source set to `call`. Example: "John Smith - New Case".

Win/Loss Tracking

Deals that reach a terminal stage are considered closed:

- **Won/Completed:** Deals that reach the industry's completed stage (e.g., "won", "completed", "checked_out"). These contribute to revenue metrics.
- **Lost/Cancelled:** Deals that reach the industry's cancelled stage (e.g., "lost", "cancelled"). These contribute to loss metrics.
- **No-Show:** For industries that track no-shows (medical, dental, restaurant, salon), a separate terminal stage captures these.

Deal Fields Reference

Field	Type	Description
`name`	Text	Deal name (required)
`description`	Text	Additional details
`company`	Text	Associated company
`stage`	Text	Current pipeline stage
`sort_index`	SmallInt	Position within column
`amount_cents`	Integer	Deal value in cents
`expected_close`	Date	Expected close date
`contact_id`	UUID	Linked contact
`call_id`	UUID	Originating call
`source`	Enum	call, web, manual, import
`created_by`	Text	Creator identifier
`archived_at`	Timestamp	Soft delete timestamp

---

8. Deal Pipeline Configuration

Industry-Specific Pipeline Stages

Lumentra adapts the pipeline stages to match your industry. Each industry has a unique set of stages that reflect the natural workflow of that business type.

Default (Generic B2B) Pipeline

Used when no specific industry is configured or for general business-to-business workflows.

Stage	Color	Terminal
New	Blue	No
Contacted	Cyan	No
Qualified	Amber	No
Proposal	Purple	No
Negotiation	Orange	No
Won	Green	Yes
Lost	Red	Yes

- **Default Stage:** new
- **Completed Stage:** won
- **Cancelled Stage:** lost

Medical and Dental Pipeline

Focused on patient appointment lifecycle with insurance verification support.

Stage	Color	Terminal
Inquiry	Blue	No
Scheduled	Cyan	No
Confirmed	Amber	No
Completed	Green	Yes
No-Show	Orange	Yes
Cancelled	Red	Yes

- **Default Stage:** inquiry
- **Completed Stage:** completed
- **Cancelled Stage:** cancelled

Additional Task Types:

- Insurance Verification
- Prescription Refill (medical only)

Restaurant Pipeline

Tracks the reservation lifecycle from initial inquiry through seating.

Stage	Color	Terminal
Inquiry	Blue	No
Reserved	Cyan	No
Confirmed	Amber	No
Seated	Purple	No
Completed	Green	Yes
No-Show	Orange	Yes
Cancelled	Red	Yes

- **Default Stage:** inquiry
- **Completed Stage:** completed
- **Cancelled Stage:** cancelled

Additional Task Types:

- Vendor Order
- Event Setup

Hotel and Motel Pipeline

Follows the guest stay lifecycle from booking through checkout.

Stage	Color	Terminal
Inquiry	Blue	No
Reserved	Cyan	No
Checked In	Amber	No
Checked Out	Green	Yes
Cancelled	Red	Yes
No-Show	Orange	Yes

- **Default Stage:** inquiry

- **Completed Stage:** checked_out
- **Cancelled Stage:** cancelled

Salon Pipeline

Tracks the appointment lifecycle for hair salons, spas, and beauty services.

Stage	Color	Terminal
Inquiry	Blue	No
Booked	Cyan	No
Confirmed	Amber	No
Completed	Green	Yes
No-Show	Orange	Yes
Cancelled	Red	Yes

- **Default Stage:** inquiry
- **Completed Stage:** completed
- **Cancelled Stage:** cancelled

Auto Service Pipeline

Tracks vehicle service orders from initial quote through completion.

Stage	Color	Terminal
Inquiry	Blue	No
Quoted	Cyan	No
Scheduled	Amber	No
In Progress	Purple	No
Completed	Green	Yes
Cancelled	Red	Yes

- **Default Stage:** inquiry
- **Completed Stage:** completed
- **Cancelled Stage:** cancelled

Additional Task Types:

- Parts Order
- Vehicle Pickup

Stage Validation

When a deal is moved to a new stage (via drag-and-drop or API), the system validates that the target stage exists in the tenant's industry pipeline. If the stage is invalid, the API returns a 400 error with a message listing all valid stages.

Deals with stages from a different industry (for example, after an industry configuration change) are collected into an "other" bucket in the pipeline view.

Pipeline View Data Structure

The pipeline API (`GET /api/deals/pipeline`) returns a stages array where each entry contains:

- `stage`: The stage identifier string
- `count`: Number of deals in this stage
- `total_amount_cents`: Sum of all deal amounts in this stage

- `deals`: Array of deal objects with joined contact names

---

9. Task Management

What Are Tasks?

Tasks are actionable items that track follow-ups, callbacks, reminders, and other work linked to contacts and deals. Tasks can be created manually or generated automatically by the post-call automation system.

Navigating to Tasks

Select **Tasks** from the sidebar navigation. The tasks page displays a data table with summary counts at the top.

Task Summary Counts

The page header shows four real-time metrics retrieved from `GET /api/tasks/counts`:

- **Pending:** Number of tasks not yet completed (done_at is null)
- **Overdue:** Number of past-due tasks (done_at is null AND due_date is before today). Highlighted in red if greater than zero.
- **Due Today:** Number of tasks due today (done_at is null AND due_date equals today)
- **Done This Week:** Number of tasks completed in the current week (done_at is not null AND done_at is after the start of the current week). Highlighted in green.

Task List View

The tasks table shows the following columns:

Column	Description
Checkbox	Click to mark done
Title	Task title (strikethrough if done)
Type	Type badge (Follow Up, Call Back)
Contact	Linked contact name or dash
Due Date	Relative date with overdue in red
Priority	Badge: Urgent, High, Medium, Low
Source	Manual, Voice Call, AI Agent

Filtering Tasks

The tasks page provides three filter controls:

Status Tabs:

- **All** – Show all tasks
- **Pending** – Uncompleted tasks (done_at IS NULL)
- **Overdue** – Past-due uncompleted tasks (done_at IS NULL AND due_date < today)
- **Done** – Completed tasks (done_at IS NOT NULL)

Type Filter (dropdown):

The available task types depend on your industry:

Universal types (all industries):

Type	Label
`follow_up`	Follow Up
`call_back`	Call Back
`email`	Email
`meeting`	Meeting
`review`	Review
`custom`	Custom

Industry-specific types:

Industry	Additional Types
Medical	Insurance Verification, Rx Refill
Dental	Insurance Verification
Restaurant	Vendor Order, Event Setup
Auto Service	Parts Order, Vehicle Pickup

Priority Filter (dropdown):

- All Priorities
- Urgent (red badge)
- High (orange badge)
- Medium (amber badge)
- Low (blue badge)

Sorting Tasks

Click the **Title** or **Due Date** column header to sort. By default, tasks are sorted by due date ascending (most urgent first). The API supports sorting by: created_at, updated_at, title, type, priority, due_date, due_time, done_at, source.

Creating a Task

1. Click the **Add Task** button in the top-right corner.
2. Fill in the task form:
    - **Title** (required): A clear description of the task.
    - **Type** (required, default: Follow Up): Select from the type dropdown.
    - **Priority** (required, default: Medium): Urgent, High, Medium, or Low.
    - **Due Date** (required, default: tomorrow): When the task should be completed.
    - **Due Time** (optional): Specific time of day for the task.
    - **Description** (optional): Additional context.
    - **Contact** (optional): Link to a contact by providing their ID.
    - **Deal** (optional): Link to a deal by providing its ID.
    - **Assigned To** (optional): User ID or name of the responsible person.
3. Click **Add Task** to save.

Completing a Task

Click the checkbox next to a task to mark it as done. The completion happens optimistically – the UI updates immediately, and if the server update fails, the change is reverted.

Completed tasks show:

- A green checkmark icon replacing the empty checkbox
- Strikethrough text on the title
- Grayed-out priority badge

- "Done" in place of the due date

The completion sets `done_at` to the current timestamp. There is no "undo" in the UI, but the task can be reopened by setting `done_at` back to null via the API.

Editing a Task

Tasks can be edited through the task form by updating any field: title, type, priority, due date, due time, description, contact link, deal link, or assignee.

Deleting a Task

Tasks are hard-deleted (permanently removed from the database), unlike contacts and deals which use soft deletion. This is done via `DELETE /api/tasks/:id`.

Viewing Upcoming and Overdue Tasks

The API provides two specialized views:

- **Upcoming tasks:** Tasks due within the next N days (default: 7 days). Accessible via `GET /api/tasks/upcoming?days=7`. Returns tasks sorted by due_date ascending, then due_time ascending (nulls last).
- **Overdue tasks:** All past-due uncompleted tasks. Accessible via `GET /api/tasks/overdue`. Returns tasks sorted by due_date ascending (oldest overdue first).

---

10. Task Automation and Workflows

How Tasks Are Auto-Created

The post-call automation engine creates tasks automatically based on call outcomes. These automated tasks reduce the risk of missed follow-ups and ensure that important actions are tracked even when staff are busy.

Automated Task Rules

Rule 1: Escalation Follow-Up

- **Trigger:** Call outcome is "escalation"
- **Task Created:**
  - Title: "Follow up: escalated call from [Contact Name]"
  - Type: `call_back`
  - Priority: `high`
  - Due Date: tomorrow
  - Source: `auto`
  - Created By: `auto`
  - Linked to the contact and call record

Rule 2: Missed/Short Call Callback

- **Trigger:** Call status is "missed" OR call duration is less than 10 seconds
- **Task Created:**
  - Title: "Missed call from [Contact Name]"
  - Type: `call_back`
  - Priority: `high`
  - Due Date: today
  - Source: `auto`
  - Created By: `auto`
  - Linked to the contact and call record

Task Lifecycle Workflow

The recommended workflow for managing tasks:

1. **Morning Review:** Open the Tasks page and check the Overdue count. Address overdue tasks first.
2. **Daily Execution:** Work through tasks sorted by due date. Complete each task by clicking the checkbox.
3. **End of Day:** Check the "Due Today" count. Any remaining tasks should be completed or rescheduled.
4. **Weekly Review:** Check "Done This Week" to measure productivity. Review upcoming tasks for the next week.

Task Priority Guidelines

Priority	When to Use
Urgent	Time-sensitive, must be done now
High	Important, should be done today
Medium	Normal priority, this week
Low	Nice to have, no firm deadline

Linking Tasks to Deals

Tasks can be linked to deals via the `deal_id` field. This is useful for tracking specific actions required to move a deal forward. For example:

- "Send proposal to client" linked to a deal at the "Proposal" stage
- "Verify insurance" linked to a medical case at the "Scheduled" stage
- "Order parts" linked to an auto service order at the "Scheduled" stage

To view all tasks for a specific deal, use `GET /api/tasks?deal_id={deal_id}`.

Escalation Rules for Overdue Tasks

While the system does not currently auto-escalate overdue tasks, the following manual process is recommended:

1. Filter tasks by status = "overdue" daily
2. For tasks overdue by more than 3 days, escalate to a manager
3. For tasks overdue by more than 7 days, consider whether the task is still relevant
4. Delete irrelevant tasks; reassign or reprioritize remaining ones

---

11. Call History

Navigating to Call History

Select **Calls** from the sidebar navigation. The call history page displays a comprehensive log of all calls handled by the voice agent.

Call List View

The calls table shows the following columns:

Column	Description
Direction	Inbound (green) or outbound (blue)
Caller	Name and phone number
Outcome	Badge showing the call result
Flow	Visual timeline bar
Duration	Call length in MM:SS format
When	Relative timestamp

Searching Calls

Use the search bar to find calls by caller name or phone number. The search applies a 300ms debounce and matches using ILIKE pattern matching.

Filtering by Outcome

Use the **outcome dropdown** to filter calls by result:

- All Outcomes
- Booking (or your industry's transaction label)
- Inquiry
- Support
- Escalated
- Hangup

Call Detail Panel

Click any call row to open the detail panel on the right side of the screen. The detail view includes:

Caller Information:

- Caller name and phone number (with copy-to-clipboard button)

Metadata Grid:

- **Duration:** Formatted as "Xm Ys"
- **Date:** Full date and time (e.g., "Mon, Jan 15, 2:30 PM")
- **Outcome:** Badge with outcome type
- **Direction:** Inbound or outbound badge

Summary:

The AI-generated summary of the call, describing what the caller wanted and what happened.

Detected Intents:

Tags showing what the caller intended (e.g., "booking", "hours_inquiry", "pricing").

Linked Contact:

If the call is linked to a contact, their name, email, and phone are shown in a card.

Linked Booking:

If the call resulted in a booking, the booking details are shown: type, status, date, time, and confirmation code.

Recording:

If a recording is available, an audio player is embedded for playback.

Transcript:

The full conversation transcript, displayed as a chat-style view with alternating bubbles for the AI Agent and the Caller. The transcript supports multiple formats:

- Structured JSON array with `role` and `content` fields
- String format with “AI:” and “User:” prefixes
- Wrapped object with `messages` or `transcript` keys

Call Data Fields

Field	Type	Description
`direction`	Enum	`inbound` or `outbound`
`status`	Enum	ringing, connected, completed, etc.
`caller_phone`	Text	Caller’s phone number
`caller_name`	Text	Caller’s name
`started_at`	Timestamp	When the call started
`ended_at`	Timestamp	When the call ended
`duration_seconds`	Integer	Call length in seconds
`outcome_type`	Enum	booking, inquiry, support, etc.
`summary`	Text	AI-generated call summary
`sentiment_score`	Decimal	-1 to +1
`intents_detected`	Array	List of detected intents
`recording_url`	URL	Link to recording
`transcript`	JSON	Full conversation
`contact_id`	UUID	Linked contact
`booking_id`	UUID	Linked booking

Call Outcomes

Outcome	Meaning
`booking`	Successfully made a booking
`inquiry`	Asked questions, did not book
`support`	Needed help with existing booking
`escalation`	Escalated to a human agent
`hangup`	Hung up before completing

Call Statuses

Status	Meaning
`ringing`	Call is ringing, not yet answered
`connected`	Call is in progress
`completed`	Call ended normally
`failed`	Call failed to connect
`missed`	Call was not answered

---

12. Call Analytics and Reporting

Navigating to Analytics

Select **Analytics** from the sidebar navigation. The analytics page provides visual insights into call performance and booking conversion.

Time Period Selection

Use the dropdown in the top-right corner to select the analysis period:

- Last 7 days
- Last 14 days
- Last 30 days (default)
- Last 90 days

Click **Refresh** to reload the data.

Summary Statistics

Four stat cards appear at the top of the page:

Metric	Description
Total Calls	Calls in the selected period
Bookings Made	Calls that converted to bookings
Conversion Rate	Percentage of calls to bookings
Avg Call Duration	Average length in MM:SS

Charts

Call Volume (Area Chart):

Daily call count plotted over time as a filled area chart. Shows trends in call volume and helps identify busy periods. Each data point represents a single day.

Call Outcomes (Pie Chart):

A donut chart showing the distribution of call outcomes:

- Booking: green
- Inquiry: blue
- Support: amber
- Escalation: red
- Hangup: gray

Labels show the outcome name and percentage.

Calls vs Bookings (Bar Chart):

A grouped bar chart comparing daily call volume (indigo) against daily bookings (green). Helps visualize conversion trends over time.

Peak Call Hours (Horizontal Bar Chart):

Shows the top 5 busiest hours of the day based on call volume. Displayed as a horizontal bar chart with hours on the Y-axis and call counts on the X-axis. Useful for staffing decisions and understanding when customers are most likely to call.

Period Comparison

Below the charts, a comparison section shows three metrics across three time frames (today, this week, this month):

- **Calls:** Today / This Week / This Month
- **Bookings:** Today / This Week / This Month
- **Estimated Revenue:** Today / This Week / This Month

Call Duration Metrics

The API provides average call duration based on the last 100 completed calls via `GET /api/calls/stats`. This rolling average smooths out outliers and gives a representative picture of typical call length.

Time Series Data

The analytics API (`GET /api/calls/analytics?days=30`) returns:

- **period:** The date range and number of days
- **summary:** Aggregated totals for the period
  - totalCalls
  - totalBookings
  - conversionRate (rounded to 1 decimal)
  - avgDurationSeconds
- **timeSeries:** Daily breakdown with:
  - date
  - calls
  - bookings
  - avgDuration
- **outcomes:** Outcome distribution as name/value pairs
- **peakHours:** Top 5 busiest hours with hour number and call count

---

13. Activity Tracking

Contact Activity Log

Every contact has an activity timeline that records all interactions in chronological order. View it in the **Activity** tab of the contact detail drawer.

Activity Types

The system tracks 19 distinct activity types:

Activity Type	Description
`call_received`	Inbound call from contact
`call_made`	Outbound call to contact
`booking_created`	New booking made
`booking_modified`	Booking was changed
`booking_cancelled`	Booking was cancelled
`booking_completed`	Booking was completed
`booking_no_show`	Contact did not show up
`sms_sent`	SMS sent to contact
`sms_received`	SMS received from contact
`email_sent`	Email sent to contact
`email_opened`	Contact opened an email
`email_clicked`	Contact clicked email link
`note_added`	Note was added
`tag_added`	Tag was added
`tag_removed`	Tag was removed
`status_changed`	Status was changed
`merged`	Contact was merged
`imported`	Contact was imported or created
`exported`	Contact data was exported

Activity Entry Details

Each activity entry contains:

- **activity_type:** The type identifier from the table above
- **description:** Human-readable text (e.g., "Status changed from active to vip")
- **created_at:** Timestamp of the activity
- **metadata:** Additional context as JSON (e.g., `{ "old_status": "active", "new_status": "vip" }`)
- **related_id:** ID of the related record (call, booking, note)
- **related_type:** Type of the related record (e.g., "call", "booking")
- **performed_by:** User or system identifier that triggered the activity

How Activities Are Generated

Activities are generated automatically by the contact service layer. Every significant operation on a contact triggers an `addActivity` call:

- Creating a contact logs an `imported` activity
- Updating contact status logs a `status_changed` activity
- Soft-deleting a contact logs a `status_changed` activity (to inactive)
- Adding a note logs a `note_added` activity
- Adding a tag logs a `tag_added` activity
- Removing a tag logs a `tag_removed` activity
- Merging contacts logs a `merged` activity on the primary

Viewing Activity History via API

```
GET /api/contacts/:id/history?limit=50&offset=0
```

Returns a paginated list of activities sorted by created_at descending (most recent first).

---

14. Post-Call Automation Engine

Overview

The post-call automation engine runs after every call is logged. It is a non-blocking process – the automation is called with `.catch()` error handling so it never blocks the call log API response. If any rule fails, the error is logged but the system continues operating normally.

Automation Context

When the automation runs, it receives the following context about the call:

- `tenantId` – The tenant that owns this call
- `callId` – The ID of the call that just completed
- `contactId` – The ID of the linked contact (if any)
- `callerPhone` – The caller's phone number
- `callerName` – The caller's name (if identified)
- `outcomeType` – The call outcome (booking, inquiry, support, escalation, hangup)
- `durationSeconds` – How long the call lasted
- `status` – The call status (completed, missed, etc.)
- `industry` – The tenant's industry configuration

If no contact is linked to the call (contactId is null), the automation exits immediately without running any rules.

Rule 1: Booking Outcome

Trigger: `outcomeType === "booking"`

Actions:

1. Creates a deal at the industry's completed/won stage
    - Name: "[Contact Name] - [Deal Label]" (e.g., "Jane Doe - Reservation")
    - Stage: The industry's completedStage (e.g., "won", "completed", "checked_out")
    - Source: "call"
    - Linked to the contact and call
    - Created by: "auto"
2. Updates the contact's lead_status to "converted" (if not already)

Early Return: After this rule fires, no further rules are evaluated.

Rule 2: Escalation Outcome

Trigger: `outcomeType === "escalation"`

Actions:

1. Creates a follow-up task
    - Title: "Follow up: escalated call from [Contact Name]"
    - Type: `call_back`
    - Priority: `high`
    - Due Date: tomorrow
    - Source: "auto"
    - Linked to the contact and call

Early Return: After this rule fires, no further rules are evaluated.

Rule 3: Missed or Very Short Call

Trigger: `status === "missed"` OR `durationSeconds < 10`

Actions:

1. Creates a callback task
    - Title: "Missed call from [Contact Name]"
    - Type: `call_back`
    - Priority: `high`
    - Due Date: today (same day)
    - Source: "auto"
    - Linked to the contact and call

Early Return: After this rule fires, no further rules are evaluated.

Rule 4: VIP Upgrade

Trigger: All of the following must be true:

- Contact has 3 or more total calls
- Contact's engagement score is 80 or higher
- Contact's current status is not already "vip"

Actions:

1. Updates the contact's status to "vip"

No Early Return: This rule can fire alongside Rule 5.

Rule 5: First-Time Inquiry

Trigger: All of the following must be true:

- Contact has 1 or fewer total calls (first-time caller)
- The call outcome is "inquiry"

Actions:

1. Creates a deal at the industry's default/initial stage
    - Name: "[Contact Name] - New [Deal Label]" (e.g., "John Smith - New Case")
    - Stage: The industry's defaultStage (e.g., "new", "inquiry")
    - Source: "call"
    - Linked to the contact and call
    - Created by: "auto"
2. Confirms the contact's lead_status as "new" (if not already set)

Rule Execution Order

Rules are evaluated in order (1 through 5). Rules 1, 2, and 3 use early returns, meaning only one of these will fire per call. Rules 4 and 5 can both fire if their conditions are met.

Error Handling

If any rule throws an error, the error is caught and logged to the console with the prefix `[AUTOMATION]`. The remaining rules are not executed for that call. This ensures that a failure in one rule does not affect the overall system stability.

---

15. Contact Segmentation and Filtering

Advanced Search Capabilities

The contact search system supports a wide range of filters that can be combined to create precise segments of your customer base.

Available Filter Parameters

Text Search:

- `search` – Matches against name, email, and phone using ILIKE pattern matching. Supports partial matches.

Status Filters:

- `status` – One or more of: active, inactive, blocked, vip, churned. Comma-separated for multiple values.
- `lead_status` – One or more of: new, contacted, qualified, converted, lost. Comma-separated for multiple values.

Source Filters:

- `source` – One or more of: call, booking, import, manual, sms, web. Comma-separated for multiple values.

Tag Filters:

- `tags` – One or more tag strings, comma-separated. Uses array overlap matching, meaning a contact matches if they have ANY of the specified tags.

Behavioral Filters:

- `has_bookings` – true/false. When true, only contacts with total_bookings > 0 are returned.
- `has_calls` – true/false. When true, only contacts with total_calls > 0 are returned.

Date Range Filters:

- `created_after` – ISO 8601 date string. Only contacts created after this date.
- `created_before` – ISO 8601 date string. Only contacts created before this date.
- `last_contact_after` – ISO 8601 date string. Only contacts whose last interaction was after this date.
- `last_contact_before` – ISO 8601 date string. Only contacts whose last interaction was before this date.

Segmentation Examples

High-Value Active Customers:

```
GET /api/contacts?status=vip,active&has_bookings=true
```

New Leads from This Month:

```
GET /api/contacts?lead_status=new&created_after=2026-03-01
```

Dormant Contacts (No Activity in 60 Days):

```
GET /api/contacts?last_contact_before=2025-12-31
```

Contacts from Phone Calls Who Never Booked:

```
GET /api/contacts?source=call&has_bookings=false
```

VIP Contacts Tagged as "Event Booking":

```
GET /api/contacts?status=vip&tags=event-booking
```

Sorting Options

Results can be sorted by any of these fields:

- `name` (alphabetical)
- `created_at` (newest or oldest first)
- `last_contact_at` (most or least recent)
- `engagement_score` (highest or lowest first)
- `total_calls` (most or fewest)
- `total_bookings` (most or fewest)
- `lifetime_value_cents` (highest or lowest)

Default sort: `last_contact_at` descending.

Pagination

All search results are paginated. The response includes:

- `data` – Array of contact records
- `total` – Total number of matching contacts
- `limit` – Records per page (default 20, max 100)
- `offset` – Current offset
- `has_more` – Boolean indicating if more records exist

---

16. Bulk Operations

Bulk Tagging

Add a tag to multiple contacts simultaneously:

```
POST /api/contacts/bulk/tags
{
  "contact_ids": ["uuid1", "uuid2", "uuid3"],
  "tag": "campaign-2026-q1"
}
```

The response returns the count of contacts updated. This uses an optimized database function (`bulk_add_tag`) for efficient batch processing.

Bulk Import

Import up to 10,000 contacts in a single operation:

```
POST /api/contacts/import
{
  "records": [
    { "phone": "+15551234567", "name": "John" },
    { "phone": "+15559876543", "name": "Jane" }
  ],
  "skip_duplicates": true,
  "update_existing": false
}
```

See Section 21 for full import documentation.

Bulk Export

Export contacts matching any filter combination:

```
POST /api/contacts/export
{
  "format": "csv",
  "filters": {
    "status": "active",
    "has_bookings": true
  }
}
```

The export respects all active filters. Supports JSON and CSV formats.

Contact Merging

Merge multiple duplicate contacts into a single primary record:

```
POST /api/contacts/merge
{
  "primary_id": "uuid-of-primary-contact",
  "secondary_ids": ["uuid-dup-1", "uuid-dup-2"]
}
```

The merge operation is described in detail in Section 3 under “Merging Duplicate Contacts”.

Data Cleanup Recommendations

Finding Duplicates:

1. Export all contacts to CSV
2. Sort by phone number to identify duplicates
3. Review contacts with similar names but different phone numbers
4. Use the merge API to consolidate duplicates

Cleaning Inactive Records:

1. Filter contacts with status “inactive” and no calls in the last 180 days
2. Review each contact to confirm they are truly inactive
3. Consider permanently deleting if no useful data exists
4. Alternatively, add a “cleanup-candidate” tag for later review

Standardizing Tags:

1. Export contacts and review the unique tag values
2. Identify inconsistent tags (e.g., “VIP” vs “vip” vs “V.I.P.”)
3. Use the tag add/remove API to standardize
4. Document the approved tag taxonomy for your team

---

17. CRM Dashboard Widgets

Overview

The Lumentra dashboard provides several widget areas that display CRM data at a glance. These widgets give business owners quick access to the most important metrics without navigating to individual pages.

Call Statistics Widget

Displays three key call metrics:

- **Calls Today:** Number of calls received today
- **Calls This Week:** Cumulative weekly call count
- **Calls This Month:** Cumulative monthly call count
- **Avg Duration:** Average call duration based on the last 100 completed calls

Data source: `GET /api/calls/stats`

Recent Calls Widget

Shows the most recent calls in a compact list format:

- Caller name (or “Unknown Caller”)
- Phone number
- Duration
- Outcome badge
- Call summary (truncated)
- Timestamp

Displays up to 10 recent calls. Data source: `GET /api/calls/recent?limit=10`

Task Counts Widget

Displays four task metrics in colored cards:

- **Pending** – Total uncompleted tasks
- **Overdue** – Past-due tasks (red highlight)
- **Due Today** – Tasks due today
- **Done This Week** – Tasks completed this week (green highlight)

Data source: `GET /api/tasks/counts`

Pipeline Summary Widget

Shows the deal pipeline at a glance:

- Total number of active deals
- Total pipeline value
- Deal count per stage
- Value per stage

Data source: `GET /api/deals/pipeline`

Outcome Distribution Widget

A summary of call outcomes for the current period:

- Booking count and percentage
- Inquiry count and percentage
- Support count and percentage
- Escalation count and percentage
- Hangup count and percentage

Data source: Derived from `GET /api/calls/analytics`

Conversion Rate Trend Widget

Shows the conversion rate (calls to bookings) as a percentage with a trend indicator showing whether it is rising or falling compared to the previous period.

---

18. Integration with Voice Agent

How Calls Feed into the CRM

The Lumentra voice agent (built on LiveKit Agents with Python) communicates with the CRM API through internal HTTP endpoints. The integration flow is:

Step 1: Contact Resolution

When a call arrives, the agent sends a POST request to `POST /api/contacts/find-or-create` with the caller's phone number. The API:

1. Normalizes the phone number to E.164 format
2. Checks the in-memory contact cache (target: <50ms)
3. If not cached, queries the database
4. If not found, creates a new contact with source "call"
5. Returns the full contact record

Step 2: Call Processing

During the call, the voice agent:

- Builds a system prompt that includes the contact's history, preferences, and booking data
- Conducts the conversation using STT (Deepgram nova-3) + LLM (GPT-4.1-mini) + TTS (Cartesia Sonic-3)
- Detects intents and tracks the conversation flow
- Executes tool calls (check_availability, create_booking, etc.) against the API

Step 3: Call Logging

When the call ends, the agent logs the call via the internal API with:

- Direction (inbound/outbound)
- Caller phone and name
- Duration in seconds
- Call status (completed, missed, failed)
- Outcome type (booking, inquiry, support, escalation, hangup)
- AI-generated summary
- Sentiment score
- Detected intents
- Full transcript
- Contact ID and booking ID (if applicable)

Step 4: Post-Call Automation

After the call is logged, the API triggers the post-call automation engine (see Section 14). This happens asynchronously and does not block the call logging response.

Voice Agent Configuration Impact on CRM

The agent's behavior directly affects what data enters the CRM:

- **Agent personality settings** affect how the agent interacts with callers, which impacts sentiment scores and outcomes
- **Call duration limits** affect how long calls can run before being terminated, which impacts the duration metric
- **Escalation configuration** determines when calls are escalated, affecting the ratio of escalation outcomes
- **Booking tool availability** determines whether callers can book during calls, affecting conversion rates

Data Latency

The CRM is updated in near-real-time:

- Contact lookup: <50ms (cached), <200ms (database)
- Call logging: <500ms
- Post-call automation: 1-3 seconds after call ends
- Engagement score recalculation: 1-5 seconds
- Notification queuing: 1-2 seconds

---

19. Notification System

Navigating to Notifications

Select **Notifications** from the sidebar navigation. The notifications page manages the notification queue, templates, and settings.

Notification Queue

The **Queue** tab displays all notifications with columns:

Column	Description
Status	Clock, spinner, check, or X icon
Channel	Envelope (email) or bubble (SMS)
Type	"Booking Confirmation", etc.
Recipient	Name and contact info
Message	Preview (truncated)

Column	Description
Created	When created
Sent	When sent

Status Filter Buttons:

- All
- Pending
- Sent
- Failed

Notification Types

Type	Description
`booking_confirmation`	Sent when booking is created
`booking_reminder_24h`	Sent 24 hours before
`booking_reminder_1h`	Sent 1 hour before
`booking_modified`	Sent when booking changes
`booking_cancelled`	Sent when booking cancelled
`booking_rescheduled`	Sent when rescheduled
`missed_call_followup`	Sent after missed call
`thank_you`	After completed booking
`review_request`	Request for review
`marketing`	Marketing (requires opt-in)
`custom`	Custom one-off messages

Notification Statuses

Notifications progress through a lifecycle:

```
pending --> queued --> sending --> sent --> delivered
                         |
                      failed --> (retry) --> sent
```

Status	Description
`pending`	Created, not yet queued
`queued`	In the delivery queue
`sending`	Currently being sent
`sent`	Sent to provider
`delivered`	Confirmed delivered
`opened`	Recipient opened (email)
`clicked`	Recipient clicked link
`bounced`	Delivery bounced
`failed`	Failed (retries up to 3x)
`cancelled`	Cancelled before sending

Notification Templates

The **Templates** tab displays configured notification templates as cards with template name, channel badge, type label, and body preview.

Default Templates:

1. **Booking Confirmation SMS:** "Hi {{customer_name}}, your {{booking_type}} is confirmed for {{booking_date}} at {{booking_time}}. Confirmation: {{confirmation_code}}. Reply CANCEL to cancel."
2. **Booking Confirmation Email:** Full HTML email with booking details, date, time, and confirmation code.
3. **Booking Reminder 24h SMS:** "Reminder: Your {{booking_type}} is tomorrow at {{booking_time}}..."
4. **Booking Reminder 1h SMS:** "Reminder: Your {{booking_type}} is in 1 hour..."
5. **Booking Cancellation SMS:** "Your {{booking_type}} for {{booking_date}} at {{booking_time}} has been cancelled..."
6. **Missed Call Follow-up SMS:** "Hi! We missed your call. How can we help?..."

Template Variables:

Templates use `{{variable_name}}` syntax. Available variables:

- `customer_name`
- `booking_type`
- `booking_date`
- `booking_time`
- `confirmation_code`
- `business_name`

Notification Settings

The **Settings** tab configures per-tenant notification preferences:

Booking Confirmations:

- Enable/disable email notifications
- Enable/disable SMS notifications

Booking Reminders:

- Configure reminder timing (default: 24 hours and 1 hour before)
- Enable/disable by channel

Review Requests:

- Enable/disable post-booking review request messages

Retry Logic

Failed notifications are retried up to 3 times (`max_retries` field). Each retry is scheduled with exponential backoff via the `next_retry_at` timestamp. After all retries are exhausted, the notification remains in "failed" status for manual review.

---

20. Escalation Management

How Escalation Works

Lumentra's voice agent includes a smart escalation manager that handles caller requests to speak with a human. The system is designed to demonstrate the AI's capabilities before escalating, while always respecting the caller's autonomy.

Immediate Escalation Triggers

The following keywords cause immediate escalation:

- **Complaints/Refunds:** “complaint”, “refund”, “money back”
- **Emergencies:** “emergency”, “urgent”, “crisis”, “medical”
- **Authority Requests:** “manager”, “supervisor”, “owner”, “boss”
- **Serious Concerns:** “harassment”, “discrimination”, “safety”
- **Legal:** “lawyer”, “lawsuit”, “sue”, “legal”
- **Account Actions:** “cancel all”, “close account”
- **Authorities:** “police”, “authorities”

Graduated Deflection Flow

Request Number	System Behavior
1st request	Soft deflection
2nd request	Escalate if valid reason, else try once
3rd request	Escalate if valid reason, else final try
4th+ request	Always escalate

Valid Escalation Reasons

Reason	Trigger Phrases
Customer Complaint	“complaint”, “complain”
Refund Request	“refund”, “money back”
Service Issue	“problem”, “issue”, “wrong”
Billing Issue	“billing”, “charge”
Cancellation	“cancel appointment”
Reschedule	“reschedule”, “change time”
Special Request	“special request”

AI Failure Escalation

The system also escalates when the AI cannot resolve the issue:

- **3+ consecutive failures:** If the AI fails three times in a row
- **10+ turns without resolution:** If the conversation reaches 10 turns without completing the task

Call Outcome for Escalations

When a call is escalated, the outcome is recorded as `escalation`. This triggers:

1. A high-priority `call_back` task due tomorrow
2. The task is linked to the contact and call record

Escalation Metrics

Each call tracks:

- Number of human requests during the call
- Total conversation turns
- Whether escalation occurred
- The reason for escalation (if applicable)

---

21. Data Import and Export

Importing Contacts

To import contacts in bulk:

1. Prepare a JSON file with an array of contact records.
2. Use the **Import** button on the contacts page (or call `POST /api/contacts/import` directly).

Import Format:

Each record must include a `phone` field. All other fields are optional:

Field	Type	Required
`phone`	String	Yes
`email`	String	No
`name`	String	No
`first_name`	String	No
`last_name`	String	No
`company`	String	No
`tags`	String or Array	No
`notes`	String	No
`custom_fields`	Object	No

When tags are provided as a string, they are split by comma and trimmed.

Import Options:

Option	Default	Description
`skip_duplicates`	false	Skip existing phone numbers
`update_existing`	false	Update existing records instead

Import Limits:

- Maximum 10,000 records per import operation.

Import Processing:

Each record is processed sequentially:

1. Phone number is validated and normalized
2. The system checks if a contact with this phone already exists
3. Based on options, the contact is created, updated, or skipped
4. Validation errors are collected with row numbers

Import Results:

The import returns a summary:

- `total`: Number of records processed
- `created`: Number of new contacts created
- `updated`: Number of existing contacts updated
- `skipped`: Number of duplicates skipped
- `errors`: Array of validation errors with row number, field, and message

Exporting Contacts

Click the **Export** button on the contacts page to export contacts.

Export Formats:

Format	Description
JSON	Full contact records with all fields
CSV	Spreadsheet-compatible format

CSV Export Columns:

- id, phone, email, name, first_name, last_name
- company, status, lead_status, tags
- total_calls, total_bookings
- lifetime_value_cents, created_at

Exports respect any active filters. To export all contacts, clear all filters before exporting. The export supports up to 100,000 records.

Import Best Practices

1. **Validate phone numbers** before importing. Invalid formats will cause row-level errors.
2. **Use skip_duplicates** on first import to avoid creating duplicate records.
3. **Use update_existing** for subsequent imports to refresh data.
4. **Break large files** into batches of 5,000-10,000 records for reliability.
5. **Include tags** during import to enable immediate segmentation.
6. **Review the error report** after every import and correct problematic rows.

---

22. Data Privacy and GDPR Compliance

Communication Preference Flags

Every contact has four compliance-related boolean flags:

- `do_not_call` – When true, the system should not initiate phone calls to this contact
- `do_not_sms` – When true, SMS notifications are suppressed
- `do_not_email` – When true, email notifications are suppressed
- `marketing_opt_in` – When true (with recorded timestamp), the contact has consented to marketing communications

Marketing Opt-In Tracking

When `marketing_opt_in` is set to true, the system automatically records the timestamp in `marketing_opt_in_at`. This provides an auditable record of when consent was given. The timestamp is not cleared if opt-in is later revoked – only the boolean flag changes.

Soft Deletion

Contacts are never permanently deleted through the standard API. The delete operation sets the status to “inactive”, preserving all associated data. This ensures:

- Audit trails remain intact
- Historical analytics are not affected
- Data can be restored if needed
- Regulatory retention requirements are met

Data Export for Subject Access Requests

To fulfill a data subject access request (DSAR), use the export functionality:

1. Search for the contact by phone or email

2. Export their complete contact record (JSON format)
3. Export their call history via `GET /api/contacts/:id/calls`
4. Export their booking history via `GET /api/contacts/:id/bookings`
5. Export their notes via `GET /api/contacts/:id/notes`
6. Export their activity history via `GET /api/contacts/:id/history`

Data Minimization

The CRM only collects data that is directly relevant to business operations:

- Phone number (required for call identification)
- Name, email, company (optional, provided by the contact)
- Call transcripts and summaries (generated from voice interactions)
- Booking details (created by the contact's explicit request)

Multi-Tenant Isolation

As described in Section 2, all data is strictly isolated by tenant. PostgreSQL Row-Level Security policies and application-level WHERE clauses ensure that one tenant's data is never accessible to another tenant.

Audit Logging

The system maintains audit logs for sensitive operations:

- Contact creation, update, and deletion
- Status changes (including VIP upgrades)
- Tag additions and removals
- Contact merges
- Data exports and imports

Each audit entry records the action, resource type, resource ID, old values, new values, user ID, IP address, and user agent.

Retention Considerations

Business owners should establish their own data retention policies. The Lumentra CRM provides the tools to:

- Identify inactive contacts (filter by last_contact_before)
- Export data before cleanup
- Soft-delete contacts that are no longer needed
- Review audit logs for compliance reporting

---

23. Industry-Specific CRM Workflows

Hotel and Motel Workflow

Typical Call Flow:

1. Guest calls to inquire about room availability
2. Voice agent checks availability and quotes rates
3. If guest books, a reservation is created
4. CRM creates a "Booking" deal at the "Reserved" stage
5. Confirmation notification sent via SMS

Pipeline Stages Used:

- Inquiry -> Reserved -> Checked In -> Checked Out
- OR: Inquiry -> Reserved -> Cancelled / No-Show

Common Tags: “returning-guest”, “corporate”, “group-booking”, “loyalty-member”

Key Metrics:

- Average booking value
- No-show rate
- Repeat guest percentage

Restaurant Workflow

Typical Call Flow:

1. Diner calls to make a reservation
2. Voice agent checks table availability for the requested date and party size
3. Reservation is created with party size and any special requests
4. CRM creates a “Reservation” deal at the “Reserved” stage
5. Confirmation notification sent

Pipeline Stages Used:

- Inquiry -> Reserved -> Confirmed -> Seated -> Completed
- OR: Reserved -> Cancelled / No-Show

Common Tags: “large-party”, “private-event”, “dietary-restrictions”, “regular”

Key Metrics:

- Covers per day
- No-show rate
- Average party size

Additional Task Types:

- Vendor Order (for special ingredients or supplies)
- Event Setup (for private dining or events)

Medical Practice Workflow

Typical Call Flow:

1. Patient calls to schedule an appointment
2. Voice agent verifies patient information
3. Agent checks provider availability
4. Appointment is created
5. CRM creates a “Case” deal at the “Scheduled” stage
6. Confirmation notification sent

Pipeline Stages Used:

- Inquiry -> Scheduled -> Confirmed -> Completed
- OR: Scheduled -> Cancelled / No-Show

Common Tags: “new-patient”, “insurance-verified”, “referral”, “follow-up-needed”

Key Metrics:

- New patient acquisition rate
- No-show rate
- Average visits per patient

Additional Task Types:

- Insurance Verification

- Prescription Refill

Dental Practice Workflow

Similar to medical, with the same pipeline stages and the addition of Insurance Verification as a task type. Common additional tags include “cleaning-due”, “treatment-plan-pending”, “orthodontics”.

Salon and Spa Workflow

Typical Call Flow:

1. Client calls to book a service (haircut, color, massage, etc.)
2. Voice agent checks stylist/therapist availability
3. Appointment is created with the requested service
4. CRM creates an “Appointment” deal at the “Booked” stage
5. Reminder notifications sent 24h and 1h before

Pipeline Stages Used:

- Inquiry -> Booked -> Confirmed -> Completed
- OR: Booked -> Cancelled / No-Show

Common Tags: “color-client”, “bridal-party”, “membership”, “first-visit”

Key Metrics:

- Client retention rate
- Average service value
- Rebooking rate

Auto Service Workflow

Typical Call Flow:

1. Customer calls about vehicle service needs
2. Voice agent collects vehicle information and service requirements
3. Agent provides a quote if possible
4. Service appointment is scheduled
5. CRM creates a “Service Order” deal at the “Quoted” or “Scheduled” stage

Pipeline Stages Used:

- Inquiry -> Quoted -> Scheduled -> In Progress -> Completed
- OR: Quoted -> Cancelled

Common Tags: “fleet-account”, “warranty”, “collision-repair”, “maintenance-plan”

Key Metrics:

- Average repair order value
- Service-to-quote conversion rate
- Repeat customer rate

Additional Task Types:

- Parts Order (for parts that need to be special-ordered)
- Vehicle Pickup (reminder to notify customer for pickup)

---

24. Advanced Reporting and KPIs

Key Performance Indicators

The following KPIs can be derived from the CRM data:

Call Performance KPIs:

- **Total Call Volume:** Total calls per day/week/month. Source: `GET /api/calls/stats`
- **Average Call Duration:** Mean duration of completed calls. Source: `GET /api/calls/stats`
- **Conversion Rate:** Percentage of calls resulting in bookings. Source: `GET /api/calls/analytics`
- **Escalation Rate:** Percentage of calls escalated to human. Calculated from outcome distribution.
- **Hangup Rate:** Percentage of calls where caller hung up. Calculated from outcome distribution.

Customer Engagement KPIs:

- **New Contacts per Period:** Count of contacts created in a date range. Use `created_after` and `created_before` filters.
- **VIP Contact Count:** Count of contacts with status "vip". Use `status=vip` filter.
- **Average Engagement Score:** Mean engagement score across all active contacts.
- **Engagement Level Distribution:** Count of contacts per engagement level (cold, warm, hot, vip).

Pipeline KPIs:

- **Total Pipeline Value:** Sum of amount_cents for all active deals. Available from the pipeline API.
- **Average Deal Size:** Mean deal amount across active deals.
- **Win Rate:** Deals reaching completed stage / total deals created.
- **Average Time to Close:** Mean days from deal creation to completed stage.
- **Stage Conversion Rates:** Percentage of deals moving from one stage to the next.

Task Performance KPIs:

- **Task Completion Rate:** Tasks completed on time / total tasks.
- **Overdue Task Count:** Number of past-due tasks. Source: `GET /api/tasks/overdue`
- **Average Tasks per Contact:** Total tasks / total active contacts.

Building Custom Reports

To build custom reports, combine the following API endpoints:

1. Use `GET /api/calls/analytics?days=N` for time-series call data
2. Use `GET /api/contacts?[filters]` for contact segmentation
3. Use `GET /api/deals?[filters]` for deal analysis
4. Use `GET /api/tasks/counts` for task performance
5. Use `GET /api/contacts/export` to extract raw data for external analysis

Peak Hours Analysis

The analytics API provides peak hours data showing the top 5 busiest hours of the day. This information is valuable for:

- Scheduling staff during high-call-volume periods
- Planning marketing campaigns to avoid peak times
- Setting voice agent capacity expectations

Sentiment Tracking

Each call includes a `sentiment_score` between -1.0 and +1.0. The contact's `sentiment_average` is a rolling average across all their calls. Use this to:

- Identify unhappy customers before they churn
- Recognize opportunities for upselling with positive-sentiment contacts

- Track sentiment trends across your customer base

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25. API Reference for CRM Operations

Base URL

All API endpoints are relative to your Lumentra API base URL. Authentication is required via session token, and the `X-Tenant-ID` header must be included in every request.

Contacts API

List/Search Contacts:

`GET /api/contacts`

Query parameters: search, status, lead_status, tags, source, has_bookings, has_calls, created_after, created_before, last_contact_after, last_contact_before, limit, offset, sort_by, sort_order

Lookup by Phone:

`GET /api/contacts/lookup?phone=+15551234567`

Returns `{ found: true, contact: {...} }` or `{ found: false, contact: null }`

Lookup by Email:

`GET /api/contacts/lookup/email?email=jane@example.com`

Get Contact:

`GET /api/contacts/:id`

Create Contact:

`POST /api/contacts`

Body: phone (required), email, name, first_name, last_name, company, source, source_details, tags, notes, custom_fields

Update Contact:

`PUT /api/contacts/:id`

Body: email, name, first_name, last_name, company, status, lead_status, preferred_contact_method, preferred_contact_time, preferred_language, timezone, do_not_call, do_not_sms, do_not_email, marketing_opt_in, tags, notes, custom_fields

Delete Contact (soft):

`DELETE /api/contacts/:id`

Find or Create by Phone:

`POST /api/contacts/find-or-create`

Body: phone (required), plus optional contact fields

Update Status:

`PATCH /api/contacts/:id/status`

Body: status (active, inactive, blocked, vip, churned)

Manage Tags:

`PATCH /api/contacts/:id/tags`

Body: `{ "add": ["tag1"], "remove": ["tag2"] }`

Bulk Add Tag:

`POST /api/contacts/bulk/tags`

Body: `{ "contact_ids": [...], "tag": "tag-name" }`

Get Notes:

`GET /api/contacts/:id/notes?limit=20&offset=0`

Add Note:

`POST /api/contacts/:id/notes`

Body: content (required), note_type, call_id, booking_id, is_pinned, is_private

Get Activity History:

`GET /api/contacts/:id/history?limit=50&offset=0`

Get Bookings:

`GET /api/contacts/:id/bookings?limit=20&offset=0`

Get Calls:

`GET /api/contacts/:id/calls?limit=20&offset=0`

Import Contacts:

`POST /api/contacts/import`

Body: records (array), skip_duplicates, update_existing

Export Contacts:

`POST /api/contacts/export`

Body: format ("json" or "csv"), filters (ContactFilters object)

Merge Contacts:

`POST /api/contacts/merge`

Body: primary_id (UUID), secondary_ids (array of UUIDs)

Recalculate Engagement Score:

`POST /api/contacts/:id/recalculate-score`

Get Engagement Breakdown:

`GET /api/contacts/:id/engagement`

Deals API

List/Search Deals:

`GET /api/deals`

Query parameters: search, stage, contact_id, source, start_date, end_date, limit, offset, sort_by, sort_order

Get Pipeline (Kanban):

`GET /api/deals/pipeline`

Returns stages array with deals grouped by stage.

Get Deal:

`GET /api/deals/:id`

Create Deal:

`POST /api/deals`

Body: name (required), description, company, stage, amount_cents, expected_close, contact_id, call_id, source, created_by

Update Deal:

`PUT /api/deals/:id`

Body: name, description, company, stage, amount_cents, expected_close, contact_id

Update Stage (Kanban drag):

`PATCH /api/deals/:id/stage`

Body: stage (required), sort_index (optional)

Archive Deal:

`DELETE /api/deals/:id`

Tasks API

List/Search Tasks:

`GET /api/tasks`

Query parameters: status, type, priority, contact_id, deal_id, start_date, end_date, limit, offset, sort_by, sort_order

Get Task Counts:

`GET /api/tasks/counts`

Returns pending, overdue, due_today, completed_this_week.

Get Upcoming Tasks:

`GET /api/tasks/upcoming?days=7`

Get Overdue Tasks:

`GET /api/tasks/overdue`

Get Task:

`GET /api/tasks/:id`

Create Task:

`POST /api/tasks`

Body: title (required), due_date (required), description, type, priority, due_time, contact_id, deal_id, call_id, assigned_to, created_by, source

Update Task:

`PUT /api/tasks/:id`

Body: title, description, type, priority, due_date, due_time, contact_id, deal_id, assigned_to

Complete Task:

`PATCH /api/tasks/:id/complete`

Delete Task:

`DELETE /api/tasks/:id`

Calls API

List Calls:

`GET /api/calls`

Query parameters: status, outcome, start_date, end_date, search, limit (max 100), offset

Get Call Statistics:

`GET /api/calls/stats`

Returns callsToday, callsWeek, callsMonth, avgDurationSeconds, outcomes.

Get Call Analytics:

`GET /api/calls/analytics?days=30`

Returns period, summary, timeSeries, outcomes, peakHours.

Get Recent Calls:

`GET /api/calls/recent?limit=10`

Get Call Details:

`GET /api/calls/:id`

Returns call record with linked contact and booking.

Get Call Transcript:

`GET /api/calls/:id/transcript`

Returns transcript and summary.

HTTP Status Codes

Code	Meaning
200	Success
201	Created
400	Validation failed
404	Resource not found
409	Conflict (duplicate phone)
500	Internal server error

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26. Industry-Specific Terminology

Lumentra adapts its terminology to match your business type. Throughout the dashboard, labels change based on your configured industry:

Concept	Default	Medical	Restaurant
Deal	Deal	Case	Reservation
Deals	Deals	Cases	Reservations
Booking	Booking	Appointment	Reservation
Customer	Contact	Patient	Guest

Concept	Hotel	Salon	Auto Service
Deal	Booking	Appointment	Service Order
Deals	Bookings	Appointments	Service Orders
Booking	Reservation	Appointment	Service
Customer	Guest	Client	Customer

This terminology is applied consistently across:

- Page titles and headers
- Table column headers (e.g., “Reservations” instead of “Bookings”)
- Form labels and placeholders
- Empty state messages
- Pipeline stage names
- Task type labels
- Deal creation forms
- Notification templates

Appendix A: Complete Field Reference

Contact Fields

Field	Type	Default
`id`	UUID	Auto-generated
`tenant_id`	UUID	Required
`phone`	Text	Required, unique/tenant
`phone_normalized`	Text	Auto (E.164)
`email`	Text	null
`name`	Text	null
`first_name`	Text	null
`last_name`	Text	null
`company`	Text	null
`source`	Enum	‘call’
`source_details`	JSONB	{}
`first_contact_at`	Timestamp	NOW()
`last_contact_at`	Timestamp	null
`last_booking_at`	Timestamp	null
`last_call_at`	Timestamp	null
`total_calls`	Integer	0
`total_bookings`	Integer	0
`total_completed_bookings`	Integer	0
`total_cancelled_bookings`	Integer	0
`total_no_shows`	Integer	0
`total_sms_sent`	Integer	0
`total_emails_sent`	Integer	0
`lifetime_value_cents`	Integer	0
`average_booking_value_cents`	Integer	0
`engagement_score`	Integer	0
`sentiment_average`	Decimal	null
`status`	Enum	‘active’
`lead_status`	Enum	null
`preferred_contact_method`	Enum	null
`preferred_contact_time`	Text	null
`preferred_language`	Text	‘en’

Field	Type	Default
timezone	Text	null
do_not_call	Boolean	false
do_not_sms	Boolean	false
do_not_email	Boolean	false
marketing_opt_in	Boolean	false
marketing_opt_in_at	Timestamp	null
custom_fields	JSONB	{}
tags	Text[]	{}
notes	Text	null
avatar_url	Text	null

Contact Source Values:

- call – Created during a phone call
- booking – Created during a booking
- import – Created via bulk import
- manual – Created manually via dashboard
- sms – Created via SMS interaction
- web – Created via web form or chat

Contact Status Values:

- active – Normal active contact
- inactive – Soft-deleted or archived
- blocked – Blocked from communications
- vip – High-value customer
- churned – Disengaged customer

Lead Status Values:

- new – First-time lead, not yet contacted
- contacted – Business has reached out
- qualified – Confirmed as a genuine prospect
- converted – Completed a booking/purchase
- lost – Did not convert

Deal Fields

Field	Type	Default
id	UUID	Auto-generated
tenant_id	UUID	Required
name	Text	Required
description	Text	null
company	Text	null
stage	Text	Industry default
sort_index	SmallInt	0
amount_cents	Integer	0
expected_close	Date	null
contact_id	UUID	null
call_id	UUID	null
source	Enum	‘manual’
created_by	Text	null
archived_at	Timestamp	null

Deal Source Values:

- `call` – Created from a phone call (auto or manual)
- `web` – Created from a web interaction
- `manual` – Created manually via dashboard
- `import` – Created via data import

Task Fields

Field	Type	Default
`id`	UUID	Auto-generated
`tenant_id`	UUID	Required
`title`	Text	Required
`description`	Text	null
`type`	Text	‘follow_up’
`priority`	Enum	‘medium’
`due_date`	Date	Required
`due_time`	Time	null
`done_at`	Timestamp	null
`contact_id`	UUID	null
`deal_id`	UUID	null
`call_id`	UUID	null
`assigned_to`	Text	null
`created_by`	Text	null
`source`	Enum	‘manual’

Task Priority Values:

- `low` – No firm deadline, nice to have
- `medium` – Normal priority
- `high` – Important, should be done soon
- `urgent` – Time-sensitive, do now

Task Source Values:

- `manual` – Created by a human user
- `auto` – Created by post-call automation
- `voice_agent` – Created by the voice agent during a call

Call Fields

Field	Type	Default
`id`	UUID	Auto-generated
`tenant_id`	UUID	Required
`vapi_call_id`	Text	null
`direction`	Enum	Required
`status`	Enum	Required
`caller_phone`	Text	null
`caller_name`	Text	null
`started_at`	Timestamp	null
`ended_at`	Timestamp	null
`duration_seconds`	Integer	null
`outcome_type`	Enum	null
`summary`	Text	null
`sentiment_score`	Decimal	null
`intents_detected`	Text[]	null
`recording_url`	Text	null

Field	Type	Default
transcript	JSONB	null
metadata	JSONB	null
contact_id	UUID	null
booking_id	UUID	null

Call Direction Values:

- `inbound` – Call received by the voice agent
- `outbound` – Call initiated by the system

Call Status Values:

- `ringing` – Call is ringing, not yet answered
- `connected` – Call is in progress
- `completed` – Call ended normally
- `failed` – Call failed to connect
- `missed` – Call was not answered

Call Outcome Values:

- `booking` – Caller made a booking
- `inquiry` – Caller asked questions
- `support` – Caller needed help
- `escalation` – Call was escalated
- `hangup` – Caller hung up

Notification Fields

Field	Type	Default
id	UUID	Auto-generated
tenant_id	UUID	Required
contact_id	UUID	null
channel	Enum	Required
notification_type	Enum	Required
status	Enum	‘pending’
recipient	Text	Required
recipient_name	Text	null
subject	Text	null
body	Text	Required
body_html	Text	null
template_id	UUID	null
template_variables	JSONB	{}
scheduled_at	Timestamp	null
sent_at	Timestamp	null
delivered_at	Timestamp	null
opened_at	Timestamp	null
clicked_at	Timestamp	null
booking_id	UUID	null
call_id	UUID	null
provider	Text	null
provider_message_id	Text	null
error_message	Text	null
retry_count	Integer	0
max_retries	Integer	3
next_retry_at	Timestamp	null

Contact Activity Fields

Field	Type	Default
id	UUID	Auto-generated
tenant_id	UUID	Required
contact_id	UUID	Required
activity_type	Enum	Required
description	Text	null
metadata	JSONB	{}
related_id	UUID	null
related_type	Text	null
performed_by	Text	null
created_at	Timestamp	NOW()

Contact Note Fields

Field	Type	Default
id	UUID	Auto-generated
tenant_id	UUID	Required
contact_id	UUID	Required
note_type	Enum	‘general’
content	Text	Required
attachments	JSONB	[]
created_by	Text	null
created_by_name	Text	null
call_id	UUID	null
booking_id	UUID	null
is_pinned	Boolean	false
is_private	Boolean	false
created_at	Timestamp	NOW()

Booking Fields (with CRM Extensions)

Field	Type	Default
id	UUID	Auto-generated
tenant_id	UUID	Required
call_id	UUID	null
contact_id	UUID	null
resource_id	UUID	null
slot_id	UUID	null
customer_name	Text	Required
customer_phone	Text	Required
customer_email	Text	null
booking_type	Text	Required
booking_date	Date	Required
booking_time	Time	Required
duration_minutes	Integer	null
notes	Text	null
status	Enum	‘pending’
confirmation_code	Text	Auto-generated
amount_cents	Integer	null
reminder_sent	Boolean	false

Field	Type	Default
`reminder_sent_at`	Timestamp	null
`source`	Enum	‘call’
`rescheduled_from`	UUID	null
`rescheduled_count`	Integer	0

Appendix B: Engagement Score Calculation Examples

Example 1: New Contact (Score: 25)

A contact who called once yesterday and made no bookings:

Factor	Raw Metric	Raw Score	Weight	Contribution
Recency	1 day	100	25	25.0
Booking Freq	0 bookings	0	20	0.0
Call Freq	1 call	25	15	3.75
Conversion	0%	0	15	0.0
Loyalty	N/A	0	15	0.0
Lifetime Val	$0	0	10	0.0

Total Score: 29 (Warm level, borderline Cold)

Example 2: Active Customer (Score: 65)

A contact who called 3 times in the last month, booked twice in 90 days, with 1 completed booking:

Factor	Raw Metric	Raw Score	Weight	Contribution
Recency	3 days	100	25	25.0
Booking Freq	2 bookings	66	20	13.2
Call Freq	3 calls	75	15	11.25
Conversion	66%	100	15	15.0
Loyalty	50% show	50	15	7.5
Lifetime Val	$150	40	10	4.0

Total Score: 76 (Hot level)

Example 3: VIP Customer (Score: 89)

A long-term customer who calls weekly, books regularly, always shows up, and has high lifetime value:

Factor	Raw Metric	Raw Score	Weight	Contribution
Recency	2 days	100	25	25.0
Booking Freq	4 bookings	100	20	20.0
Call Freq	5 calls	100	15	15.0
Conversion	80%	100	15	15.0
Loyalty	95% show	95	15	14.25
Lifetime Val	$2,500	100	10	10.0

Total Score: 99 (VIP level)

This contact would also be automatically upgraded to VIP status by the post-call automation (3+ calls AND score >= 80).

Example 4: Dormant Contact (Score: 8)

A contact who called once 4 months ago, never booked, and has not interacted since:

Factor	Raw Metric	Raw Score	Weight	Contribution
Recency	120 days	0	25	0.0
Booking Freq	0 bookings	0	20	0.0
Call Freq	0 recent	0	15	0.0
Conversion	0%	0	15	0.0
Loyalty	N/A	0	15	0.0
Lifetime Val	$0	0	10	0.0

Total Score: 0 (Cold level)

Note: The call frequency is 0 because it measures calls in the last 30 days, not all-time calls.

Example 5: High-Value but Infrequent (Score: 52)

A contact who books expensive services but only once every 2 months:

Factor	Raw Metric	Raw Score	Weight	Contribution
Recency	45 days	40	25	10.0
Booking Freq	1 booking	33	20	6.6
Call Freq	0 recent	0	15	0.0
Conversion	100%	100	15	15.0
Loyalty	100% show	100	15	15.0
Lifetime Val	$800	80	10	8.0

Total Score: 55 (Warm level)

This example shows that even high-value customers can have moderate scores if they do not interact frequently. The recency and call frequency factors naturally penalize infrequent interaction.

Appendix C: Glossary

Archived Deal: A deal that has been soft-deleted by setting the `archived_at` timestamp. Archived deals are excluded from pipeline views and searches.

Booking: A confirmed appointment, reservation, or service order created by a customer through the voice agent or manually by staff. Terminology varies by industry.

Bulk Operation: An action performed on multiple records simultaneously, such as adding a tag to multiple contacts or importing many contacts at once.

Call Outcome: The result classification of a completed call. One of: booking, inquiry, support, escalation, or hangup.

Churned Contact: A contact who has stopped engaging with the business, typically identified by a low engagement score and no recent interactions.

Contact: A record representing a person who has interacted with the business, identified primarily by their phone number.

Contact Activity: An audit log entry recording a specific action or event related to a contact, such as a status change or tag addition.

Contact Note: A free-form text record attached to a contact, used to document preferences, observations, complaints, or follow-up items.

Conversion Rate: The percentage of calls that result in a booking outcome. Calculated as (booking outcomes / total calls) * 100.

Custom Fields: A JSONB column on the contact record that allows storing arbitrary business-specific data as key-value pairs.

Deal: A business opportunity in the sales pipeline. Terminology varies by industry (Case, Reservation, Booking, Appointment, Service Order).

Deal Pipeline: The visual representation of deals organized by stage, displayed as a Kanban board.

Deflection: The voice agent's attempt to resolve a caller's request before escalating to a human agent. Uses graduated approach.

E.164 Format: The international phone number format starting with a plus sign followed by the country code and number (e.g., +15551234567).

Engagement Level: A classification derived from the engagement score: cold (0-29), warm (30-59), hot (60-79), or vip (80-100).

Engagement Score: A numeric value from 0 to 100 that quantifies how engaged a contact is with the business, based on six weighted factors.

Escalation: The process of transferring a caller from the AI voice agent to a human representative.

Hard Delete: Permanent removal of a record from the database. Used for tasks. Contacts and deals use soft deletion instead.

ILIKE: A PostgreSQL operator for case-insensitive pattern matching, used in search queries.

Industry Pipeline: The set of deal stages configured for a specific industry type (medical, dental, restaurant, hotel, salon, auto_service, or default).

Intent: A purpose or goal detected from the caller's speech, such as "booking", "hours_inquiry", or "pricing".

Kanban Board: A visual project management tool that displays deals as cards organized into columns representing pipeline stages.

Lead Status: A classification indicating where a contact is in the sales funnel: new, contacted, qualified, converted, or lost.

Lifetime Value: The total revenue generated by a contact, stored in cents in the `lifetime_value_cents` field.

Multi-Tenant: A software architecture where a single instance serves multiple customers (tenants) with strict data isolation between them.

Normalized Phone: A phone number converted to a standardized format (E.164) for consistent lookup and deduplication.

Optimistic Update: A UI pattern where the interface updates immediately after a user action, without waiting for the server response. If the server rejects the change, the UI reverts.

Pagination: The practice of dividing large result sets into pages, controlled by `limit` and `offset` parameters.

Post-Call Automation: A set of rules that execute automatically after every call is logged, creating deals, tasks, and updating contact information.

RLS (Row-Level Security): A PostgreSQL feature that restricts which rows are visible to database queries based on policies, used for tenant isolation.

Recency Decay: The natural reduction in engagement score over time as the recency factor decreases when a contact stops interacting.

Sentiment Score: A numeric value from -1.0 (very negative) to +1.0 (very positive) assigned to each call based on the caller's tone and language.

Show Rate: The percentage of bookings that were completed (customer showed up) versus total bookings. Used in the loyalty factor of engagement scoring.

Soft Delete: Marking a record as inactive or archived without removing it from the database. Used for contacts (status set to inactive) and deals (archived_at set).

Sort Index: A numeric value that determines the display order of deals within a pipeline stage column.

Tag: A free-form text label attached to a contact for categorization and filtering purposes.

Task: An actionable item representing work to be done, such as a follow-up call, email, or meeting.

Tenant: A business or organization using the Lumentra platform. Each tenant has isolated data and configuration.

Terminal Stage: A pipeline stage that represents a closed deal (won, lost, completed, cancelled, no-show). Deals in terminal stages are excluded from active pipeline value calculations.

Transcript: The full text record of a voice call conversation, stored as structured JSON with role and content fields for each utterance.

VIP: A contact status indicating a high-value customer. Can be set automatically by the post-call automation engine (3+ calls AND score >= 80) or manually by staff.

Voice Agent: The AI-powered phone system that answers calls, conducts conversations, and performs actions (bookings, lookups) on behalf of the business.

This guide covers all CRM functionality available in Lumentra as of March 2026. For deployment and infrastructure documentation, refer to the LiveKit Deployment Status guide. For voice agent configuration, refer to the Agent Setup guide.